

The Convex Origin of Fixed Costs

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TWO TRADITIONS OF ADJUSTMENT COSTS

The central puzzle: micro lumpiness vs. macro smoothness.

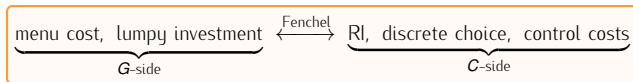
- ▶ **Fixed cost tradition:** threshold rules, binary lumpy adjustment — menu costs, lumpy investment.
- ▶ **Convex cost tradition:** smooth continuous actions — quadratic adjustment costs, rational inattention.
- ▶ **Sharply different macro predictions:** monetary/inaction non-neutrality, Phillips curve slope, welfare costs.

Are these genuinely different economics, or two views of the same object?

Micro lumpiness vs. Macro smoothness

Fixed-cost and convex probability-cost models are **ex-ante equivalent** (Fenchel); in the menu-cost case the dual coincides with Alvarez-Lippi-Oskolkov (2022) §2.2 (henceforth ALO). The contribution is not the duality but its **four consequences**:

- (i) **Closed-form** $G \mapsto C$, **environment-free**. A single construction ports across applications — no re-derivation per model.
- (ii) **Differentiability** $\Rightarrow$ **smooth FOC**. Value-function smoothness is inherited from the dual cost; threshold machinery is replaced by $S = C'(\varphi)$.
- (iii) **Rosetta stone: one G , five literatures**.



- (iv) **Operational payoff**. Fully **global, non-linear** solution of the Calvo–Goloso–Lucas spectrum via Repeated Transition Method — delivering **state-dependent monetary transmission** conditioned on the cross-sectional dispersion of price gaps (out of reach for linearised methods).

Calibration. $G(\xi) = (\xi/\bar{\xi})^\gamma$ spans Calvo ($\gamma \rightarrow 0$)–Goloso–Lucas ($\gamma \rightarrow \infty$); structural mapping $\nu = 2\gamma$ anchors γ to ALO's hazard exponent: $\nu \approx 2 \Rightarrow \hat{\gamma} \approx 1$, selection share $\approx 67\%$.

THE DUALITY

LUMPY ADJUSTMENT PROBLEM

Start from the simplest concrete instance — a Uniform menu cost. Consider an extensive-margin decision where adjustment entails a fixed cost $\xi \sim \text{Unif}(0, \bar{\xi})$:

$$v = \int_0^{\bar{\xi}} \max\{v^a - \xi, v^n\} \left(\frac{1}{\bar{\xi}}\right) d\xi, \quad v^a > v^n.$$

The optimal solution is a stochastic threshold rule: if ξ is low enough, adjust; otherwise, do not adjust. The threshold equates the cost of the marginal adjuster to the gain $v^a - v^n$:

$$\text{Adjust} \iff \xi \leq \xi^* := \min\{v^a - v^n, \bar{\xi}\}.$$

The probability of adjustment is then $\mathbb{P}(\xi \leq \xi^*) = \xi^* / \bar{\xi} = \min\left\{\frac{v^a - v^n}{\bar{\xi}}, 1\right\}$.

Two objects emerge: the threshold ξ^ and the realised probability of adjustment.*

PROBABILITY CHOICE PROBLEM

Now re-cast the same decision in a different language: instead of *drawing* a cost, directly *choose* the adjustment probability p and pay a convex penalty $c(p) = \frac{1}{2}\bar{\xi} p^2$:

$$v = \max_{p \in [0,1]} \left\{ p v^a + (1-p) v^n - \frac{1}{2}\bar{\xi} p^2 \right\}, \quad v^a > v^n.$$

The optimality condition equates the marginal benefit of raising p to the marginal cost $\bar{\xi} p$:

$$v^a - v^n = \bar{\xi} p^*, \quad p^* \in [0, 1].$$

The probability of adjustment is $p^* = \min \left\{ \frac{v^a - v^n}{\bar{\xi}}, 1 \right\}$ — identical to the threshold probability above.

$[Fenchel \text{ duality}] \quad p^* = \mathbb{P}(\xi \leq \xi^*)$

Same value, same probability. Two faces of one problem.

THEOREM 1: ANALYTICAL CONVEX DUALITY

The two formulations are equivalent under the closed-form analytical link:

$$\mathcal{C}(\varphi) = w \int_0^{G^{-1}(\varphi)} \xi \, dG(\xi)$$

- ▶ Payoff equivalence: $\Pi^{FC}(\mathbf{S}) = \Pi^{PC}(\mathbf{S})$ for all $\mathbf{S} \geq \mathbf{0}$
- ▶ Policy equivalence: $\varphi^* = G(\mathbf{S}/w)$ in both formulations
- ▶ Fenchel duality structure: Π^{PC} and $\mathcal{C}$ are Fenchel conjugates of each other
- ▶ Path-by-path identity: realised binary adjustment indicators coincide under natural coupling

The quantile integral gives the static, environment-free analytical link between G and $\mathcal{C}$.

PROOF IN THREE LINES

1. **Expanded payoff:** substitute threshold $\xi^* = S/w$ into fixed-cost payoff

$$\Pi^{FC}(S) = S \cdot G(S/w) - w \int_0^{S/w} \xi dG(\xi)$$

2. **Change variables:** let $\varphi = G(S/w)$ so $S/w = G^{-1}(\varphi)$

$$\Pi^{FC}(S) = \varphi \cdot S - w \int_0^{G^{-1}(\varphi)} \xi dG(\xi) = \varphi \cdot S - C(\varphi)$$

3. **Convexity check:** differentiating C yields $C'(\varphi) = wG^{-1}(\varphi)$ (non-decreasing)

$$C''(\varphi) = \frac{w}{g(G^{-1}(\varphi))} > 0 \quad \Rightarrow \quad C \text{ strictly convex}$$

$$\text{FOC } S = C'(\varphi) \text{ is sufficient } \Rightarrow \max_{\varphi} \{\varphi S - C(\varphi)\} = \Pi^{PC}(S)$$

Economic intuition: Next slide.

CONVEXITY IS SELECTION (NOT ASSUMPTION)

The convex-cost tradition — Rotemberg pricing, quadratic capital adjustment, RI's quadratic attention penalty — *assumes* convexity as a modelling convenience, without a micro-story for why.

Here convexity is derived, with concrete economic content:

- ▶ **Marginal cost:** $C'(\varphi) = w G^{-1}(\varphi)$ — the cost faced by the *marginal adjuster*, the agent indifferent at the current threshold:
- ▶ Raising φ requires inducing a *more expensive* agent to act $\Rightarrow$ marginal cost rises $\Rightarrow$ **convexity is selection**
- ▶ **Curvature:** $C''(\varphi) = w/g(G^{-1}(\varphi))$ — the *inverse density of marginal types*; bunched $g \Rightarrow$ gentle rise; sparse $g \Rightarrow$ sharp rise

Economics: selection makes the marginal non-adjuster more expensive than the last adjuster
 $\rightarrow$ increasing MC $\rightarrow$ convexity.

Recasts an assumed object as a derived one:

- ▶ Quadratic adjustment cost ($C \propto \varphi^2$) is what uniform G induces (Khan-Thomas)
- ▶ Power-law adjustment cost ($C \propto \varphi^{(\gamma+1)/\gamma}$) is what power-law G induces (details will follow)
- ▶ Each “functional form” for C is secretly a distributional assumption on G

Proposition (Universality). The deterministic fixed cost is a singular limit.

► **Degenerate:** $G = \delta_{\bar{\xi}}$ gives $C(\varphi) = w\bar{\xi}\varphi$ (linear, no convexity); only $\varphi \in \{0, 1\}$.

► **Any ε -perturbation:** $G_\varepsilon = U[\bar{\xi} - \varepsilon, \bar{\xi} + \varepsilon]$ gives

$$C_\varepsilon(\varphi) = w(\bar{\xi} - \varepsilon)\varphi + w\varepsilon\varphi^2, \quad C_\varepsilon''(\varphi) = 2w\varepsilon > 0.$$

► As $\varepsilon \rightarrow 0^+$: $C_\varepsilon \rightarrow w\bar{\xi}\varphi$ pointwise, $C_\varepsilon'' \rightarrow 0$, and $\varphi_\varepsilon^*(S) \rightarrow \mathbf{1}[S \geq w\bar{\xi}]$.

Reframing: the convex model is generic; the lumpy model is the singular boundary case.

DIFFERENTIABILITY THEOREM (THEOREM 2)

For dynamic lumpy adjustment problems with absolute-continuity assumptions:

- ▶ Value function: $v(x; X) \in \mathcal{C}^1$ in state x
- ▶ Intensive policy: $x'^*(x; X)$ is $\mathcal{C}^1$ everywhere
- ▶ Extensive policy: $\Lambda^*(x; X)$ is $\mathcal{C}^1$ where $\Lambda^* \in (0, 1)$
- ▶ Envelope: v_x given by Benveniste–Scheinkman

Why this matters: standard menu-cost methods need root-finding and kink handling. The duality delivers a smooth FOC $\mathbf{S} = \mathbf{C}'(\Lambda)$ with no threshold machinery — the model slots directly into FOC-based heterogeneous-agent solvers (Reiter 2009, Auclert et al. 2020).

Relation to prior work. The $\mathcal{C}^1$ regularity itself is standard, following from general dynamic-programming arguments (Santos 1991) and obtained in lumpy settings via smooth-pasting (Balei-Blanco 2021). The cost object coincides with that of ALO (2022), recovered there from the hazard. *The contribution is the route: the map from the cost distribution to the convex cost is environment-free, and it yields a closed-form FOC $\mathbf{S} = \mathbf{C}'(\varphi)$ in any environment admitting a scalar adjustment surplus.*

- ▶ Khan-Thomas (2008): firms draw heterogeneous fixed costs uniform on $[0, \bar{\xi}]$
- ▶ Duality reveals *why* firm-level investment is smooth: uniform $\mathbf{G}$ induces $\mathbf{C}(\varphi) = \frac{w_{\bar{\xi}}}{2} \varphi^2$, strictly convex with smooth FOC $\varphi^* = \mathbf{S}/(w_{\bar{\xi}})$
- ▶ *The individual firm's problem is not lumpy* — smoothness is a property of the cost distribution, not of aggregation
- ▶ Structural reading: any $\mathbf{G}$ with continuous positive density on its support produces strictly convex $\mathbf{C}$ and smooth firm-level behaviour
- ▶ GE channels (demand aggregation, factor-price feedback) are a separate matter

G AS ROSETTA STONE

The reverse direction

- ▶ *Forward* (G-side): Pick a distribution G of fixed costs $\Rightarrow$ solve for cost function C via the quantile integral

$$C(\varphi) = w \int_0^{G^{-1}(\varphi)} \xi dG(\xi)$$

- ▶ *Reverse* (C-side): Given a cost function $C(\varphi)$, what is the implied distribution G ?
- ▶ The link: $C'(\varphi) = wG^{-1}(\varphi)$ pins down G uniquely (up to scaling)
- ▶ Implication: Each literature's cost specification is an *implicit assumption* on the underlying distribution of fixed costs
 - Not a free modelling choice — an assumption that can be made explicit, tested, and recovered

THE ROSETTA STONE: FIVE LITERATURES, ONE PRIMITIVE

Distribution G	Cost $C(\varphi)$	Form	Entropy	Literature
Exponential	$(1 - \varphi) \ln(1 - \varphi) + \varphi$	Log. MC	KL divergence	Rational inattention
Power law (γ)	$\varphi^{1+1/\gamma}$	Iso-elastic	Tsallis q -div.	Generalised choice
Uniform ($\gamma = 1$)	φ^2	Quadratic	L^2	Sparsity / KT2008
Degenerate	$\bar{\xi} \varphi$ on $\{0, 1\}$	Linear	—	Lumpy adjustment
Type I EV (mv)	$\sum_j \varphi_j \ln \varphi_j$	Neg. entropy	Shannon	Logit discrete choice

- ▶ Each row: a distribution G on the left, its induced cost C via the quantile integral, and the literature where that cost function appears
- ▶ *Headline:* Five literatures, one unifying parameter family — the distribution G of physical fixed costs
- ▶ The choice of C in your model is secretly a choice of G .

From fixed to convex: closed-form welfare for menu costs

- ▶ Tractable welfare measure for fixed-cost models:

$$\text{Welfare cost} = \mathcal{C}(\varphi^*) = w \int_0^{G^{-1}(\varphi^*)} \xi \, dG(\xi)$$

- ▶ The *expected physical cost paid by adjusters* — a resource-cost reading
- ▶ For power law G : $\mathcal{C}(\varphi^*)$ closed-form alongside $\Lambda(x)$ and κ

From convex to fixed: welfare reading of information costs

- ▶ RI with exponential G : KL cost $\lambda \sum \varphi_j \ln(1/\varphi_j)$ (in nats) — λ is a shadow price of attention, no resource reading
- ▶ Fixed-cost primitive: $\mathcal{C}(\varphi^*)$ is expected physical cost; λ becomes the *rate of the underlying cost distribution*, recoverable from micro data
- ▶ Same $\mathcal{C}(\varphi)$, two structural interpretations; G selects which applies

EXTENDED DUALITY VIA EXTERNALITIES

FRICTION-EXTENDED DUALITY: $\tilde{G}$ AND $\tilde{C}$

Let an aggregate friction $F = F(X) \in (0, 1]$ multiply the adjustment case (match / success probability). An agent enters iff

$$F \cdot V^a + (1 - F) \cdot V^n - w \xi_i \geq V^n \iff F \cdot S \geq w \xi_i.$$

Aggregate participation $\varphi = G(F S / w)$. Define the *friction-extended duality*:

$$\tilde{G}(z) \equiv G(F z) \quad \text{on } [0, \bar{\xi}/F], \quad \tilde{C}(\varphi) = \frac{1}{F} C(\varphi).$$

The FOC under friction:

$$S = \tilde{C}'(\varphi) = \frac{C'(\varphi)}{F} \iff F \cdot S = C'(\varphi)$$

Power-law specialisation. $\tilde{G}$ is the *same* power-law family with shape γ unchanged and support stretched to $\bar{\xi}/F$; $\tilde{C}(\varphi) = (1/F) (w \bar{\xi}^\gamma / (\gamma + 1)) \varphi^{(\gamma+1)/\gamma}$. **Cost-shape γ is orthogonal to the friction level F .**

SEARCH-AND-MATCHING AS THE CANONICAL INSTANTIATION

In equilibrium, F is endogenous: $F = F(\theta)$ where market tightness $\theta = v/u$ depends on aggregate participation φ . Each agent takes F *parametrically*; in equilibrium it is determined jointly.

Canonical specification: Diamond-Mortensen-Pissarides matching, $M(u, v) = Au^\alpha v^{1-\alpha}$, $F(\theta) = A\theta^{1-\alpha}$.

- ▶ The matching technology supplies F as a single scalar; the duality absorbs it via $\tilde{G}, \tilde{C}$
- ▶ F depends on aggregate φ through tightness: the duality is fully compatible with general-equilibrium feedback
- ▶ Cost-shape γ and matching elasticity α remain separable margins (shape-invariance of $\tilde{G}$)

What this demonstrates: the framework reaches GE matching frictions without re-derivation. The well-known search-efficiency results — congestion externalities and the Hosios (1990) condition — follow as special cases of the construction, recovered on the next slide.

RECOVERING HOSIOS (1990) AS A SPECIAL CASE

When the friction $F = F(\theta)$ is supplied by an SaM matching technology, the friction-extended duality reproduces the canonical efficiency structure of the search literature:

$$\underbrace{F \cdot S}_{\text{private FOC}} + \underbrace{\frac{\partial F}{\partial \varphi} S \varphi}_{\text{congestion (Hosios)}} = \underbrace{C'(\varphi)}_{\text{planner FOC}} .$$

The wedge is the canonical congestion externality; the **Pigouvian entry tax** $\tau^*(\varphi) = -(\partial F / \partial \varphi) S \varphi$ decentralises the planner — pinning down Hosios (1990), generalised by Mangin-Julien (2021).

What this demonstrates:

- ▶ The dual machinery *absorbs* GE matching frictions without re-derivation
- ▶ Hosios wedge and Pigouvian tax follow as a special case of the construction
- ▶ Cost-shape γ and matching elasticity α stay orthogonal: selection and search separable

Contribution claim. Representational: the unification reaches GE matching frictions; the efficiency results recovered here are Hosios (1990) and Mangin-Julien (2021), not new claims of this paper.

PRICING APPLICATION

The **aggregate state** X : $X = (f, \text{exo. shocks})$

where $f(x)$ is the cross-sectional distribution of price gaps (following each firm's drift under inflation: $x \rightarrow x - \pi'$), and exogenous aggregates (demand, monetary policy).

The firm's value function in convex-cost form:

$$\begin{aligned} V(x; X) &= \max_{\Lambda \in [0,1]} \left\{ \Lambda V^a(X) + (1 - \Lambda) V^n(x; X) - C(\Lambda) \right\}, \\ V^a(X) &= \max_{x'} \left\{ R(x'; X) + \beta \mathbb{E} \left[m(X, X') V(x' - \pi'; X') \right] \right\}, \\ V^n(x; X) &= R(x; X) + \beta \mathbb{E} \left[m(X, X') V(x - \pi'; X') \right]. \end{aligned}$$

Three key quantities:

- ▶ $V^a(X)$: *value of repricing*—choose optimal reset gap x' , clear today's mispricing
- ▶ $V^n(x; X)$: *value of not repricing*—gap drifts by inflation $-\pi'$, future mispricing remains
- ▶ *Repricing surplus*: $S(x; X) := V^a(X) - V^n(x; X)$ — the gain from closing the gap

The Fenchel duality FOC $S = C'(\Lambda)$ pins the repricing hazard state-by-state.

REPRICING WITH STATIC APPROXIMATION

Why a static approximation? The full dynamic surplus $S(\mathbf{x}; \mathbf{X})$ is hard to compute analytically. But near the optimum, flow mispricing losses are *approximately quadratic* in the gap. This quadratic structure is also stable across the parameter space—allowing a closed-form decomposition of the Phillips slope into frequency and selection that survives recalibration.

Static repricing rule: approximate $S(\mathbf{x}; \mathbf{X}) \approx B\mathbf{x}^2$ (Alvarez-Le Bihan-Lippi 2016; ALO 2022), so repricing occurs when

$$w\tilde{\xi} \leq B\mathbf{x}^2.$$

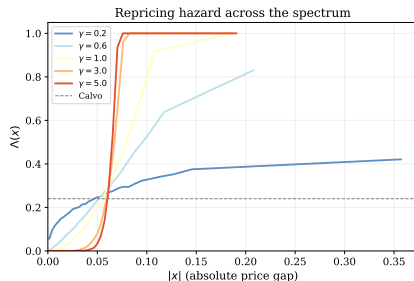
Apply the duality (Theorem 1): By Theorem 1, this fixed-cost problem maps exactly onto its convex-cost dual. The extensive-margin FOC $\mathbf{S} = \mathbf{C}'(\Lambda)$ becomes

$$\Lambda(\mathbf{x}) = G\left(\frac{B\mathbf{x}^2}{w}\right),$$

the state-dependent *repricing hazard*. For power-law $G(\xi) = (\xi/\bar{\xi})^\gamma$, this simplifies to $\Lambda(\mathbf{x}) = (B\mathbf{x}^2/(w\bar{\xi}))^\gamma$.

This hazard is the single primitive from which all aggregates follow: the Phillips slope, the selection wedge, monetary non-neutrality.

HAZARD SHAPES ACROSS THE SPECTRUM



- **One knob, two poles:** as γ rises the hazard morphs continuously from flat (Calvo) to step (Goloso–Lucas).
- Local slope $d \log \Lambda / d \log x = 2\gamma$: the chain rule reads off the selection elasticity directly from the curve.
- Intermediate γ (e.g. ALO's $\gamma = 1$) sits squarely inside the spectrum — neither extreme is a knife-edge.

THE CALVO–GOLOSOV–LUCAS SPECTRUM

Two canonical pricing models correspond to polar extremes of the hazard:

- ▶ **Calvo (1983)**: Repricing probability $\Lambda(\mathbf{x}) = \lambda$ is constant, independent of the gap. Cannot arise from any non-degenerate absolutely continuous $\mathbf{G}$.
- ▶ **Golosov–Lucas (2007)**: Fixed cost $\xi = \bar{\xi}$ is deterministic. Hazard is a step function: $\Lambda(\mathbf{x}) = \mathbf{1}[|\mathbf{x}| \geq \bar{\mathbf{x}}]$ with inaction band $\bar{\mathbf{x}} = \sqrt{w\bar{\xi}/B}$. Produces the most concentrated repricing.

Power-law family $\mathbf{G}(\xi) = (\xi/\bar{\xi})^\gamma$ spans the spectrum:

- ▶ Low γ : Dispersed costs, gentle hazard $\rightarrow$ Calvo-like
- ▶ High γ : Concentrated costs, steep hazard $\rightarrow$ Golosov–Lucas-like
- ▶ Single parameter $\gamma =$ degree of cost concentration; hazard: $\Lambda(\mathbf{x}) = \min\{(B\mathbf{x}^2/(w\bar{\xi}))^\gamma, 1\}$

MICRO-FOUNDATION OF ALO'S GENERALISED HAZARD ELASTICITY

ALO, in a continuous-time sticky-price model with Brownian gap and Poisson cost arrivals, show that a generalised hazard $\Lambda(\mathbf{x})$ admits two foundations: random menu cost $\mathbf{G}$ (their Section 2.1) and convex flow cost $\mathbf{c}(\ell)$ on Poisson intensity (their Section 2.2), with $\mathbf{c}$ recovered from Λ by solving a 2nd-order linear ODE.

Where $\mathbf{C}$ and ALO's $\mathbf{c}$ coincide: on $\varphi \in [0, 1]$, Theorem 1's $\mathbf{C}(\varphi)$ equals $\mathbf{w} \cdot \mathbf{c}(\varphi)$: both equal $\mathbf{w} \int_0^\varphi \mathbf{G}^{-1}(s) ds$.

In pricing: duality microfound the hazard elasticity.

- ▶ The duality generates the entire hazard $\Lambda(\mathbf{x}) = \mathbf{G}(\mathbf{B}\mathbf{x}^2/\mathbf{w})$ from $(\mathbf{G}, \text{surplus})$
- ▶ ALO posit a reduced-form generalised hazard with local power law $\Lambda(\mathbf{x}) - \bar{\Lambda} \sim \mathbf{c}|\mathbf{x}|^\nu$ as $\mathbf{x} \rightarrow 0$; the observed slope ν is their hazard elasticity

$$\underbrace{\nu}_{\text{how fast adjustment probability rises with the gap}} = \underbrace{2}_{\text{how fast the incentive rises with the gap}} \times \underbrace{\gamma}_{\text{how fast the adjuster pool fills with the incentive}}$$

Identification. The benefit-side “2” is pinned by theory, so the *observed* hazard exponent ν identifies the *unobserved* cost-shape γ : ALO's $\nu \approx 2 \Rightarrow \hat{\gamma} \approx 1$. A structural primitive read off a reduced-form moment.

Scope: the “2” is the local benefit curvature under the standard quadratic-surplus approximation (AL/ALO); portable, static, and environment-free across applications.

PHILLIPS CURVE DECOMPOSITION: FREQUENCY VS. SELECTION

Let $\kappa \equiv \partial\pi/\partial\mu$ be the **Phillips slope**: the elasticity of inflation to a monetary nudge μ that shifts every firm's desired price. To first order in μ , the price-level response is linear in firm-level changes, so two margins enter *additively*.

Frequency (intensive)

Repricers pass the nudge through.

$$+ \bar{\Lambda}$$

Calvo benchmark

Selection (extensive)

Marginal large-gap firms tip over threshold.

$$+ 2\gamma \bar{\Lambda}$$

Cor. 2: power-law $G \propto \xi^\gamma$, quadratic surplus

$$\kappa(G) = \underbrace{\bar{\Lambda}}_{\text{frequency}} + \underbrace{2\gamma \bar{\Lambda}}_{\text{selection}} = (1 + 2\gamma) \bar{\Lambda}$$

Selection share $= 2\gamma/(1 + 2\gamma)$ — set by cost-concentration alone; same $2\gamma = \nu$ as the previous slide.

SELECTION SHARE ACROSS THE COST-CONCENTRATION SPECTRUM

With $\kappa = (1 + 2\gamma)\bar{A}$, the *selection share* $2\gamma/(1 + 2\gamma)$ depends *only* on the tail parameter γ (not on $\bar{A}$, B , or w) — a single number indexes the entire Calvo-to-Golosov-Lucas continuum.

Model	γ	Selection share
Calvo	0	0%
Kurtosis match	0.4	44%
ALO benchmark	1	$\frac{2}{3} \approx 67\%$
Near Golosov-Lucas	4	89%
Golosov-Lucas	∞	100%

Takeaway: the empirically relevant ALO case puts **two-thirds of the Phillips slope on selection** — frequency alone (the Calvo benchmark) captures only one-third of the response to a monetary shock.

GLOBAL SOLUTION

GLOBAL SOLUTION: CONFIRMING THE ANALYTICAL DECOMPOSITION

The analytical decomposition $\kappa = (1 + 2\gamma)\bar{\Lambda}$ rests on two approximations:

- ▶ *Static surplus*: $\mathcal{S}(x) = Bx^2$ — repricing gain is quadratic in the price gap
- ▶ *Interior support*: stationary gap distribution f_{ss} sits strictly inside the inaction band, with boundary mass negligible

Purpose of the global solution: two objectives.

- ▶ **Verify** the closed-form $\kappa = (1 + 2\gamma)\bar{\Lambda}$ holds in the full dynamic GE model where neither approximation is exact (dynamic surplus only approximately quadratic, local curvature $\sim 4\times$ static; survives because Cor. 1's invariance $\kappa/\bar{\Lambda} = 1 + 2\gamma$ does not depend on B).
- ▶ **Generate state-dependent GIRFs** — *the operational payoff*: paired Monte-Carlo impulse responses conditioned on cross-sectional price-gap dispersion, delivering state-dependent monetary transmission across the spectrum that linearised methods cannot produce.

Relation to Costain-Nakov. Costain-Nakov (2011, JME) solve smoothly-state-dependent pricing computationally via Reiter linearisation. The present solution is via RTM (Lee 2026) without aggregate linearisation.

Using the *Repeated Transition Method* (Lee, 2026), I globally solve the dynamic pricing model.

What the method produces:

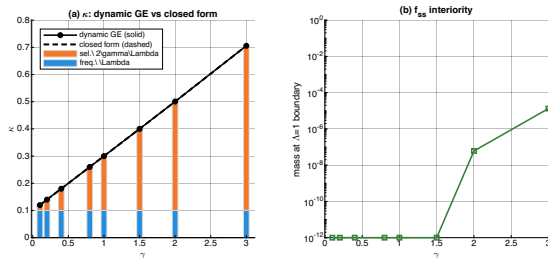
1. Solves at the benchmark and traces the spectrum across γ
2. Tracks the cross-sectional distribution f_t along its simulated path (no forecasting rule)
3. Produces aggregate dynamics without log-linearisation around steady state
4. Preserves the value-function kink at the inaction-band boundary

Two ingredients make this possible:

- ▶ Convex-cost representation $\Rightarrow$ smooth FOC without threshold machinery
- ▶ RTM matching on lagged price-dispersion Δ_{t-1} as the matching state

Note on scope. State-dependence in IRFs has prior precedent (Costain-Nakov-Petit); the dynamic exercise here serves as a consistency check on the static decomposition and as a vehicle for state-dependent IRFs at the benchmark.

SPECTRUM ACROSS γ IN THE DYNAMIC MODEL



Notes: (a) dynamic GE solution (solid) tracks the closed-form $(1 + 2\gamma)\bar{\Lambda}$ (dashed) across the spectrum. (b) Boundary mass $\leq 10^{-5}$ even at $\gamma = 3$, so the correction to κ is numerically zero.

- **Static closed-form is quantitatively accurate in the dynamic GE model:** hazard retains power-law form $\Lambda(x) = (Bx^2/(w\bar{\xi}))^\gamma$; local curvature $\approx 4\times$ static (discounted accumulation), and $\kappa = (1 + 2\gamma)\bar{\Lambda}$ holds as a tight numerical approximation.
- **Selection share preserved:** curvature-invariance $\kappa/\bar{\Lambda} = 1 + 2\gamma$ is independent of B ; range $\sim 17\%$ (low γ , Calvo-like) to $\sim 86\%$ (high γ , GL-like), with the $\frac{2}{3} \approx 67\%$ benchmark at $\gamma = 1$ intact.

STATE-DEPENDENT GIRF: COMPARATIVE STATIC ACROSS γ

RTM generates **generalised impulse responses** (paired Monte-Carlo, conditioning on lagged price-dispersion Δ_{t-1} quartiles within TFP).

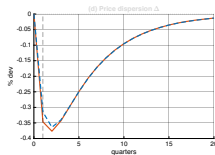
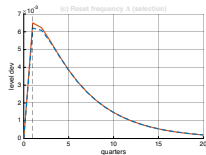
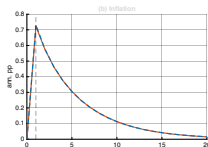
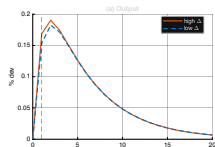
Impact response to +1-SD preference (demand) shock, by regime:

	$\hat{\gamma} = 0.4$ (sel. 44%)		$\hat{\gamma} = 0.5$ (sel. 50%)		$\hat{\gamma} = 1$ (sel. $\frac{2}{3}$)	
	hi- Δ	lo- Δ	hi- Δ	lo- Δ	hi- Δ	lo- Δ
Y (%)	+0.169	+0.154	+0.161	+0.147	+0.127	+0.120
π (ann. pp)	+0.723	+0.731	+0.692	+0.699	+0.584	+0.586
$\bar{\Delta}$ (lvl)	+0.0065	+0.0062	+0.0069	+0.0066	+0.0074	+0.0073
Δ (%)	-0.347	-0.315	-0.321	-0.294	-0.236	-0.225
Y hi/lo gap	+10.0%		+9.0%		+5.6%	

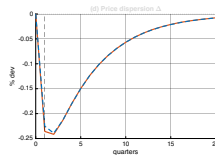
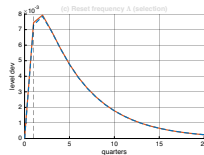
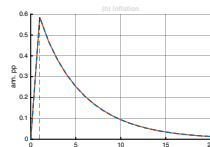
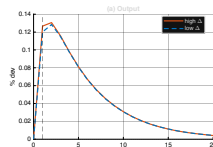
Takeaway: state-dependence is modest at all three calibrations and weakens monotonically as γ rises.

- ▶ Demand shock raises both Y and π ; selection responds on impact via $\bar{\Delta} \uparrow$ and $\Delta \downarrow$
- ▶ High- Δ output amplification: 10% $\rightarrow$ 9% $\rightarrow$ 6% as γ rises from 0.4 to 0.5 to 1
- ▶ Mechanism: as selection becomes dominant, adjustment concentrates on the most-mispriced firms regardless of cross-section composition $\Rightarrow$ less state-dependence

GIRF FIGURES: $\gamma = 0.4$ vs $\gamma = 1$



(a) $\hat{\gamma} = 0.4$ (kurtosis-matching alt.)



(b) $\hat{\gamma} = 1$ (benchmark)

- High- Δ vs low- Δ paths fan out at $\gamma = 0.4$ but nearly collapse at $\gamma = 1$.
- On impact, $\bar{\Lambda}$ jumps and pulls Δ down — the selection channel is visible, not just inferred.
- **Benchmark $\gamma = 1$ is the headline: ALO-consistent, residual state-dependence is small.**

The duality applies to **any extensive-margin decision with heterogeneous costs**:

- ▶ **Firm entry/exit**: free-entry equates marginal entrant's cost to surplus; heterogeneous entry costs G micro-found the smooth extensive margin
- ▶ **Vacancy creation**: convex costs from heterogeneous vacancy creation costs; $C''(\varphi)$ reflects the density g
- ▶ **Labour force participation**: smooth supply elasticity $\approx g(G^{-1}(\varphi))/w$, without Rogerson (1988)-type lotteries
- ▶ **Technology adoption**: hazard $\Lambda(x)$ derived from the cost distribution, not assumed
- ▶ **Credit default**: same structure when costs are drawn from G

Identification: γ from $\log \Lambda \propto 2\gamma \log |x|$; G from size-distribution moments of adjustment events.

CONCLUSION

Three central insights:

1. **Fixed and convex costs are dual representations.** The distinction is not between two kinds of economic friction but between two parametrisations of the same friction, linked by Fenchel conjugacy. The structural primitive is the **distribution G of heterogeneous fixed costs**.
2. **The Rosetta Stone.** Five literatures — menu costs, lumpy investment, discrete choice, rational inattention, control costs — each assume a shape for G . Making this assumption explicit converts a free modelling choice into a testable distributional restriction.
3. **The Calvo–Golosov–Lucas spectrum.** A single parameter γ (the power-law exponent) governs the degree of cost concentration and traces a continuous spectrum from dispersed costs (Calvo-like, weak selection) to concentrated costs (Golosov–Lucas-like, strong selection). The structural mapping $\nu = 2\gamma$ ties γ to ALO's hazard exponent. **ALO's $\nu \approx 2$ implies the benchmark $\hat{\gamma} = 1$, with selection contributing $\frac{2}{3} \approx 67\%$ of the Phillips curve slope** — selection as the dominant flexibility margin.

- ▶ $\text{fixed} \equiv \text{convex}$ — Fenchel duality
- ▶ $\nu = 2\gamma$ — benefit $\times$ cost shape
- ▶ ALO's $\nu \approx 2 \Rightarrow \hat{\gamma} = 1$; selection $\frac{2}{3}$ of PC slope
- ▶ Smooth FOC — global solvers for lumpy models

APPENDIX

▶ Geometric analysis of dynamic equilibrium

- Solow (1956); Swan (1956); Ramsey (1928); Cass (1965); Koopmans (1963)

▶ Global stochastic equilibrium solution framework

- Marcet (1988); Den Haan and Marcet (1990); Krusell and Smith (1998); Den Haan, (1996, 1997); Reiter (2001); Algan et al. (2008, 2010); Den Haan and Rendahl (2010); Reiter (2010); Ahn et al. (2018); Boppart et al. (2018); Elenev et al. (2021); Auclert et al. (2021); Cao et al. (2023); Azinovic et al. (2022); Fernández-Villaverde et al. (2023); Han et al. (2025); Payne et al. (2025); Lee (2026)

▶ Nonlinear equilibrium dynamics — state dependence

- Kaplan and Violante (2014); Vavra (2014); Berger and Vavra (2015); Basu and Bundick (2017); Bloom et al. (2018); Kaplan et al. (2018); Petrosky-Nadeau et al. (2018); Baley and Blanco (2019); Pizzinelli et al. (2020); Berger et al. (2021); Melcangi (2024); Winberry (2021); Lee (2026)

▶ Random dynamical system

- Arnold (1998); Schenk-Hoppé (1998); Schenk-Hoppé (2001); Yannacopoulos (2011)

BENCHMARK NUMBERS AT $\hat{\gamma} = 1$

Plugging the benchmark $\hat{\gamma} = 1$, $\bar{\Lambda} = 0.24$ into the closed-form decomposition $\kappa = (1 + 2\gamma)\bar{\Lambda}$:

$$\kappa = 3 \times 0.24 = 0.72$$

$$\text{Frequency component} = \bar{\Lambda} = 0.24 \quad (33\% \text{ of } \kappa)$$

$$\text{Selection wedge} = \kappa - \bar{\Lambda} = 0.48 \quad \left(\frac{2}{3} \approx 67\% \text{ of } \kappa\right)$$

Non-neutrality interpretation:

- ▶ Nominal non-neutrality: $1 - \kappa = 0.28$
- ▶ Relative to Calvo at same frequency: $(1 - \kappa)/(1 - \bar{\Lambda}) = 0.28/0.76 \approx 0.37$
- ▶ Selection reduces non-neutrality by $\approx 63\%$ relative to Calvo benchmark

Relation to Costain-Nakov. They decompose numerically across the spectrum; the closed-form selection share $2\gamma/(1 + 2\gamma)$ depending only on γ (not $\bar{\Lambda}$, B , or w) is the new layer.

THREE CANONICAL CORRESPONDENCES: DERIVATIONS

1. Exponential $G(\xi) = 1 - e^{-\lambda \xi}$

- ▶ $C(\varphi) = \frac{w}{\lambda} [(1 - \varphi) \ln(1 - \varphi) + \varphi]$; $C'(\varphi) = -\frac{w}{\lambda} \ln(1 - \varphi)$
- ▶ Information-theoretic counterpart: *generalized KL divergence*
- ▶ Adjustment: $\varphi^* = 1 - e^{-\lambda S/w}$

2. Power law $G(\xi) = (\xi/\bar{\xi})^\gamma$ on $[0, \bar{\xi}]$

- ▶ $C(\varphi) = \frac{\gamma}{\gamma+1} w \bar{\xi} \varphi^{1+1/\gamma}$ (iso-elastic, exponent $q = 1 + 1/\gamma$)
- ▶ Information-theoretic counterpart: *Tsallis q -divergence*
- ▶ γ controls cost concentration: $\gamma = 1 \rightarrow$ quadratic; $\gamma \rightarrow 0$ dispersed; $\gamma \rightarrow \infty$ concentrated

3. Uniform $G = U[0, \bar{\xi}]$ (equivalently $\gamma = 1$)

- ▶ $C(\varphi) = \frac{w \bar{\xi}}{2} \varphi^2$ (quadratic, L^2 penalty)
- ▶ Foundational in Khan-Thomas (2008) and Gabaix (2014)

POWER-LAW FAMILY: CURVATURE AND DENSITY

For $G(\xi) = (\xi/\bar{\xi})^\gamma$ with $\gamma > 0$:

$$\text{Density: } g(\xi) = \frac{\gamma}{\bar{\xi}} \left(\frac{\xi}{\bar{\xi}} \right)^{\gamma-1}$$

$$\text{Induced cost: } C(\Lambda) = \frac{\gamma}{\gamma+1} w_{\bar{\xi}} \Lambda^{1+1/\gamma}$$

Shape interpretation:

- ▶ $\gamma < 1$: density *decreasing* — cheap adjustment abundant; C more curved than quadratic
- ▶ $\gamma = 1$ (uniform): $G = U[0, \bar{\xi}]$ gives $C(\Lambda) = (w_{\bar{\xi}}/2)\Lambda^2$ — the Khan-Thomas case
- ▶ $\gamma > 1$: density *increasing* — cheap adjustment rare; C less curved than quadratic

γ alone governs the spectrum: frictionless ($\gamma \rightarrow 0$) to maximum selection ($\gamma \rightarrow \infty$).

IDENTIFYING γ : HAZARD-EXPONENT BENCHMARK

Benchmark identification (structural hazard exponent):

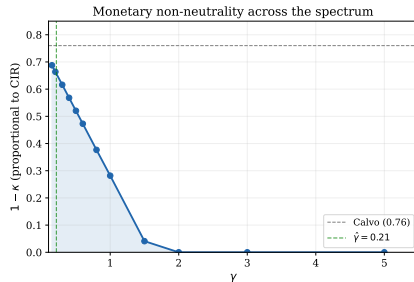
- ▶ Structural mapping (Remark after Thm. 1): $\nu = 2\gamma$ where $\Lambda(x) \propto |x|^\nu$
- ▶ ALO (2022) recover heterogeneity-corrected $\nu \approx 2$ from Cavallo scraped CPI data
- ▶ Implies $\hat{\gamma} = \nu/2 = 1$ (at the Khan-Thomas uniform- $\mathbf{G}$ point of the spectrum)

Implied benchmark calibration:

$$\hat{\gamma} = 1, \quad \kappa = 3\bar{\Lambda}, \quad \text{selection share} = \frac{2}{3}.$$

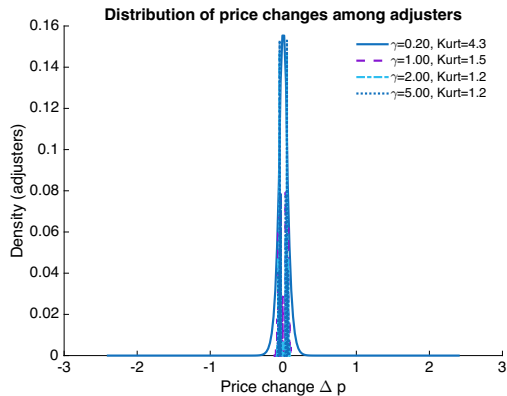
Alternative reading. Purely moment-matching against raw CPI kurtosis (heterogeneous- σ_z , $CV=1$) yields $\hat{\gamma}^{\text{kurt}} \approx 0.4$ as an alternative (selection share 44%); the structural hazard-exponent identification carries the benchmark.

CIR ACROSS THE SPECTRUM (ALI KURTOSIS IDENTITY)



- ▶ **ALI (2016) kurtosis-CIR identity:** for the power-law family, $\text{CIR}(\gamma) = \text{Kurtosis}(\gamma) / (6\bar{A})$ — the canonical reduced-form bridge from price-change moments to monetary non-neutrality.
- ▶ CIR collapses monotonically across the spectrum: Calvo ceiling at $\gamma \rightarrow 0$, Golosov–Lucas floor at $\gamma \rightarrow \infty$; at the benchmark $\hat{\gamma} = 1$ ($\nu = 2$), CIR already sits well below the Calvo ceiling.
- ▶ Connects to the empirical kurtosis-identification debate (Cavallo data, ALO's heterogeneity correction): the same kurtosis that pins down γ pins down monetary non-neutrality.

BACKUP: CALIBRATED DISTRIBUTION OF PRICE CHANGES



Heterogeneous- σ_z alternative ($\hat{\gamma} = 0.4$) reproduces the uncorrected leptokurtic CPI distribution; the benchmark $\hat{\gamma} = 1$ matches ALO's heterogeneity-corrected kurtosis ≈ 2 .

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